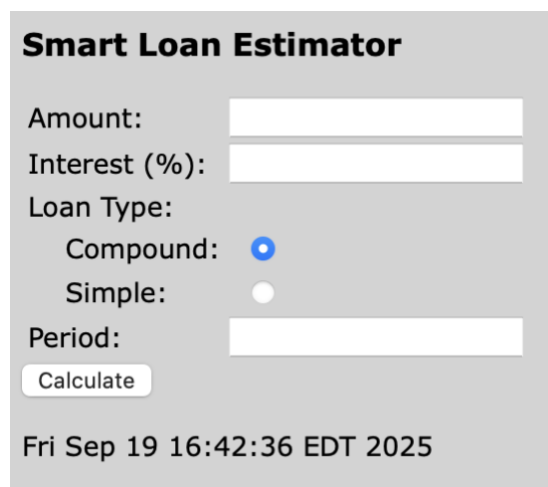


CGS4854 Website Construction and Management

Exercise 3 [40 points]

Loan Calculator using JSP

This exercise implements a Smart Loan Estimator using JSP, HTML, and server-side logic as shown in Fig. 1. The goal is to build a small web application that calculates loan repayment amounts based on Simple Interest or Compound Interest, depending on user input. The workflow of the application is shown in Fig. 2.



The image shows a web form titled "Smart Loan Estimator". It contains several input fields: "Amount:", "Interest (%):", "Loan Type:" (with radio buttons for "Compound:" and "Simple:"), and "Period:". Below these fields is a "Calculate" button. At the bottom of the form, it displays the date and time: "Fri Sep 19 16:42:36 EDT 2025".

Fig. 1. Loan Estimator Homepage

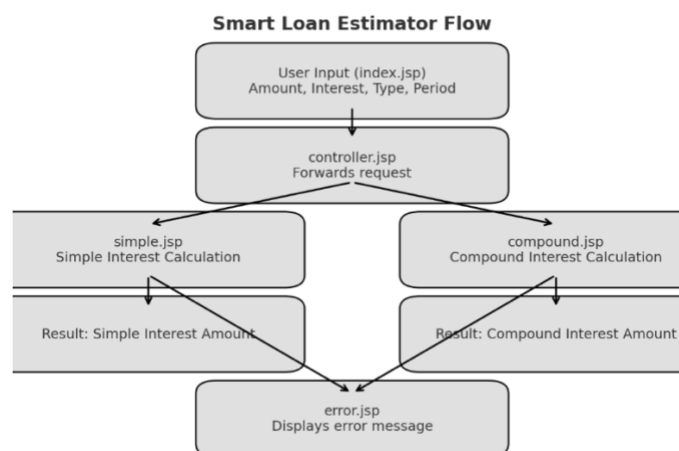


Fig.2. Flow Diagram

Steps to Create the Project in NetBeans

1. Create a New Project

- Open NetBeans.
- Go to:
File → New Project → Java with Maven/Ant → Web Application → Next
- Project Name: `loanestimatorYOURNAME`
(replace YOURNAME with your actual name, e.g., `loanestimatorERICWILLIAM`)
- Choose your server (e.g., Apache Tomcat) and click Finish.

2. Add JSP/HTML Files

Inside the project, create the following pages:

- `index.jsp` → Input form for loan calculation (Amount, Interest, Loan Type, Period).
- `controller.jsp` → Forwards request to either `simple.jsp` or `compound.jsp`.
- `simple.jsp` → Calculates loan using Simple Interest formula.
- `compound.jsp` → Calculates loan using Compound Interest formula.
- `error.jsp` → Displays error messages when invalid inputs are entered.
- `header.html` → Displays the heading Smart Loan Estimator.
- `footer.jsp` → Displays current date and time.

3. Organize Files

- Place `header.html` and `footer.jsp` in the Web Pages folder.
- All `.jsp` files (`index.jsp`, `controller.jsp`, `simple.jsp`, `compound.jsp`, `error.jsp`) should also be in the Web Pages folder.

4. Run the Project

- Right-click on the project → Select Run.
- The application will open in the browser showing the Smart Loan Estimator form.
- Enter amount, interest, type (Simple/Compound), and period → then click Calculate.
- Results will display accordingly, or error messages if inputs are invalid.

Submission Instructions

- This is an individual exercise (no group work).
- Make sure your project is named exactly: `loanestimatorYOURNAME`
- After completing and testing:
 - Go to your project folder in NetBeans workspace.
 - Compress (zip) the entire project folder.
 - Submit the zipped folder on Canvas under Exercise 2.

Deliverables

- A working JSP project with all required pages.
- The zipped project folder uploaded to Canvas.
- The program should correctly calculate **Simple** and **Compound** interest, display results, and handle errors gracefully.

Files and Their Specifications

1. **index.jsp**

- Acts as the **input form** for the user.
- Collects:
 - Loan Amount
 - Interest Rate (%)
 - Loan Type (Simple or Compound)
 - Loan Period (years)
- Submits data to `controller.jsp` for processing.
- Uses `header.html` (for title) and `footer.jsp` (for timestamp).

2. **controller.jsp**

- Controls the flow based on user choice of loan type.
- If Simple Interest is selected → forwards to `simple.jsp`.
- If Compound Interest is selected → forwards to `compound.jsp`.

3. **simple.jsp**

- Contains a method to calculate Simple Interest using the formula:

$$A = P \times \left(1 + \frac{R \times T}{100}\right)$$

- Validates that inputs are greater than 0.
- Displays the calculated principal with simple interest.
- If invalid input is given, it forwards to `error.jsp`.

4. **compound.jsp**

- Contains a method to calculate Compound Interest using the formula:

$$A = P \times \left(1 + \frac{R}{100}\right)^T$$

- Validates inputs (amount, interest, and period > 0).
- Displays the principal with compound interest.
- If invalid, forwards to `error.jsp`.
-

5. **error.jsp**

- Displays error messages if the user provides invalid inputs.
- Example errors:
 - *Amount should be greater than 0*
 - *Interest should be greater than 0*
 - *Period should be greater than 0*

6. **header.html**

- Displays the heading: "Smart Loan Estimator"

7. **footer.jsp**

- Displays the current system date and time

Source Files

index.jsp

```
<html>
  <head>
    <title>Include</title>
  </head>
  <body style="font-family:verdana;font-size:10pt;">
    <%@ include file="header.html" %>
    <form action="controller.jsp">
      <table border="0" style="font-family:verdana;font-size:10pt;">
        <tr>
          <td>Amount:</td>
          <td><input type="text" name="amount" />
        </tr>
        <tr>
          <td>Interest in %:</td>
          <td><input type="text" name="interest"/></td>
        </tr>
        <tr>
          <td>Compound:</td>
          <td><input type="radio" name="type" value="C" checked/></td>
        </tr>
        <tr>
          <td>Simple:</td>
          <td><input type="radio" name="type" value="S" /></td>
        </tr>
        <tr>
          <td>Period:</td>
          <td><input type="text" name="period"/></td>
        </tr>
      </table>
      <input type="submit" value="Calculate"/>
    </form>
    <jsp:include page="footer.jsp"/>
  </body>
</html>
```

controller.jsp

```
<%
  String type = request.getParameter("type");
  if(type.equals("S")) {
%>
<jsp:forward page="/simple.jsp"/>
<%
  } else {
%>
  <jsp:forward page="/compound.jsp"/>
<%
  }
%>
```

error.jsp

```
<%@ page isErrorPage="true" %>
<html>
  <head>
    <title>Simple</title>
  </head>
  <body style="font-family:verdana;font-size:10pt;">
    <%@ include file="header.html" %>
    <p style="color=#FF0000"><b><%= exception.getMessage() %></b></p>
    <jsp:include page="footer.jsp"/>
  </body>
</html>
```

header.html

```
<h3>Smart Loan Estimator</h3>
```

footer.jsp

```
<%= new java.util.Date() %>
```

simple.jsp

```
<%@ page errorPage="error.jsp" %>
<%!
  public double calculate(double amount, double interest, int period) {
    if(amount <= 0) {
      throw new IllegalArgumentException("Amount should be greater than 0: " + amount);
    }
    if(interest <= 0) {
      throw new IllegalArgumentException("Interest should be greater than 0: " +
interest);
    }
    if(period <= 0) {
      throw new IllegalArgumentException("Period should be greater than 0: " + period);
    }
    return amount*(1 + period*interest/100);
  }
%>
<html>
  <head>
    <title>Simple</title>
  </head>
  <body style="font-family: verdana;font-size:10pt;">
    <%@ include file="header.html" %>
    <%
      double amount = Double.parseDouble(request.getParameter("amount"));
      double interest = Double.parseDouble(request.getParameter("interest"));
      int period = Integer.parseInt(request.getParameter("period"));
    %>
    <b>Principal using simple interest:</b>
    <%= calculate(amount, interest, period) %>
    <br/><br/>
    <jsp:include page="footer.jsp"/>
  </body>
</html>
```

compound.jsp

```
<%@ page errorPage="error.jsp" %>
<%!
    public double calculate(double amount, double interest, int period) {
        if(amount <= 0) {
            throw new IllegalArgumentException("Amount should be greater than 0: " + amount);
        }
        if(interest <= 0) {
            throw new IllegalArgumentException("Interest should be greater than 0: " +
interest);
        }
        if(period <= 0) {
            throw new IllegalArgumentException("Period should be greater than 0: " + period);
        }
        return amount*Math.pow(1 + interest/100, period);
    }
%>
<html>
    <head>
        <title>Compound</title>
    </head>
    <body style="font-family:verdana;font-size:10pt;">
        <%@ include file="header.html" %>
        <%
            double amount = Double.parseDouble(request.getParameter("amount"));
            double interest = Double.parseDouble(request.getParameter("interest"));
            int period = Integer.parseInt(request.getParameter("period"));
        %>
        <b>Principal using compound interest:</b>
        <%= calculate(amount, interest, period) %>
        <br/><br/>
        <jsp:include page="footer.jsp"/>
    </body>
</html>
```

Rubric

Criteria	Excellent	Good	Satisfactory	Needs Improvement	Marks
Project Setup (5 marks)	Project correctly created in NetBeans with proper name loanestimatorYOURNAME; runs without errors.	Minor naming/structure issues but runs.	Runs with warnings or misconfigured server.	Incorrect setup; project doesn't run.	/5
Input Form (5 marks)	Form includes all required fields, well-formatted and user-friendly.	All fields included but layout could improve.	Some fields missing or poor alignment.	Form incomplete or non-functional.	/5
Controller Logic (5 marks)	Correct forwarding to simple/compound pages based on selection.	Works but has minor logic redundancy.	Works for only one case correctly.	Incorrect or missing logic.	/5
Simple Interest (5 marks)	Correct formula, validates inputs, displays accurate results.	Works but missing some validation.	Formula works but calculation errors possible.	Incorrect or missing implementation.	/5
Compound Interest (5 marks)	Correct formula, validates inputs, displays accurate results.	Works but missing some validation.	Formula works but calculation errors possible.	Incorrect or missing implementation.	/5
Error Handling (5 mark)	Clear, meaningful error messages shown for invalid inputs.	Errors displayed but messages unclear.	Errors shown inconsistently.	No error handling implemented.	/5
Reusability (5 marks)	Proper use of header.html and footer.jsp.	Both included but minor formatting issues.	Only one included.	None included.	/5
Documentation & Flow Diagram (5 marks)	Well-written description + clear flow diagram included.	Description or diagram included, but not both.	Limited explanation, unclear diagram.	No documentation provided.	/5